ALFIDO TECH JAVA INTERNSHIP

TASK 1 : JAVA BASICS & OOP FUNDAMENTALS

Objective

The objective of this task is to understand Java basics and Object-Oriented Programming concepts such as encapsulation, inheritance, polymorphism, loops, arrays, and conditional statements.

Requirements Covered

✓ Loops

✓ Conditions

✓ Arrays

✓ Classes and Objects

✓ Inheritance

✓ Polymorphism

✓ Encapsulation using Getters and Setters

OOP Concepts Used

Encapsulation

Encapsulation is achieved by making variables private and accessing them using getter methods.

Inheritance

GraduateStudent inherits Student class using extends keyword.

Polymorphism

A Student reference is used to create a GraduateStudent object.

Tools Used

- Java
- Eclipse IDE
- JDK 17

Source Code

Student.java

```
package task1;
public class Student {
   private int id;
   private String name;
   public Student(int id, String name) {
      this.id = id;
      this.name = name;
   }
   public int getId() {
      return id;
   }
   public String getName() {
      return name;
   }
   public void display() {
      System.out.println("ID: " + id);
      System.out.println("Name: " + name);
   }
}
```

GraduateStudent.java

```
package task1;
public class GraduateStudent extends Student {
   private String department;
   public GraduateStudent(int id, String name, String department) {
      super(id, name);
      this.department = department;
   }
```

```java
    @Override
    public void display() {
        super.display();
        System.out.println("Department: " + department);
    }
}
```

Main.java

```java
package task1;
public class Main {
    public static void main(String[] args) {
        int[] marks = {80, 85, 90};
        for(int mark : marks) {
            System.out.println("Mark: " + mark);
        }
        Student s1 = new GraduateStudent(101,"Shifani","CSE");
        s1.display();
        if(marks[0] > 50) {
            System.out.println("Pass");
        }
        else {
            System.out.println("Fail");
        }
    }
}
```

Screenshots

Student.java

```java
package task1;

public class Student {

    private int id;
    private String name;

    public Student(int id, String name) {
        this.id = id;
        this.name = name;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
    }
}
```

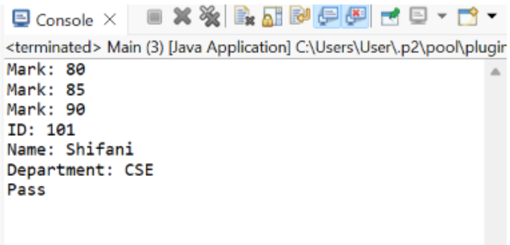
GraduateStudent.java

```java
package task1;

public class GraduateStudent extends Student {

    private String department;

    public GraduateStudent(int id, String name, String department) {
        super(id, name);
        this.department = department;
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Department: " + department);
    }
}
```

Main.java

```java
package task1;
public class Main {

    public static void main(String[] args) {

        int[] marks = {80, 85, 90};

        for(int mark : marks) {
            System.out.println("Mark: " + mark);
        }

        Student s1 = new GraduateStudent(101,"Shifani","CSE");

        s1.display();

        if(marks[0] > 50) {
            System.out.println("Pass");
        }
        else {
            System.out.println("Fail");
        }
    }
}
```

Program Output



Output

Mark: 80

Mark: 85

Mark: 90

ID: 101

Name: Shifani

Department: CSE

Pass

Conclusion

Successfully implemented Java Basics and OOP concepts including encapsulation, inheritance, polymorphism, loops, arrays, and conditions.